

Addressing the Subsumption Thesis

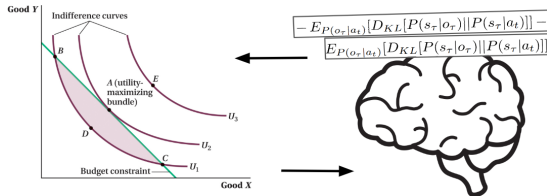
A Formal Bridge between Microeconomics and Active Inference

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Background



Subsumption Thesis

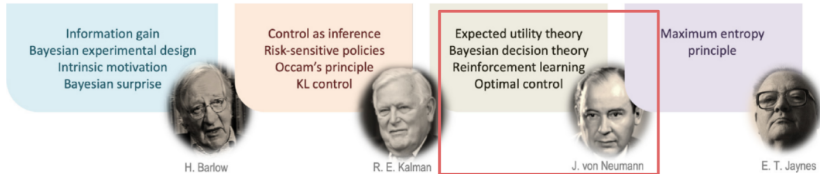


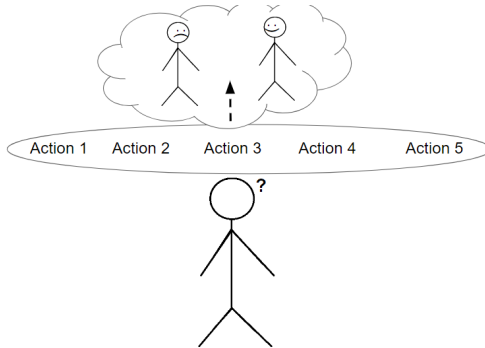
Figure 1: Active Inference subsuming other accounts of behaviour. Figure from Friston et al., 2023

“Active inference [...] englobes the principles of expected utility theory [...] it is theoretically possible to rewrite any RL algorithm [...] as an active inference algorithm.” ¹

→ Expected utility theory, as seen in economics, is subsumed by active inference – it is an edge case

Decision Theory

- First Principles of Agency
- Exploitation vs. Exploration dilemma
- Economic welfare implications



Agent-Environment Frameworks:

Commensurable Objective Functions in a

- Markov Decision Process (MDP):
 $(\mathcal{S}, \mathcal{A}, P(s'|a, s), R(s'), \gamma = 1, \mathbb{T})$
- Partially Observable MDP (POMDP):
 $(\mathcal{S}, \mathcal{A}, P(s'|a, s), P(o|a, s'), R(s'), \gamma = 1, \mathbb{T})$

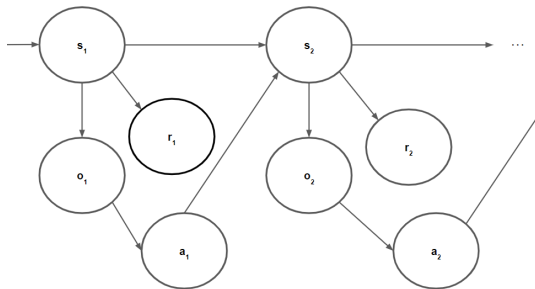
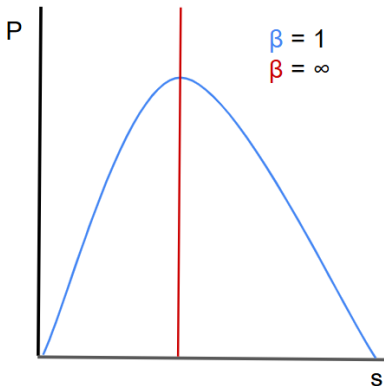


Figure 2: Graphical Model of a POMDP

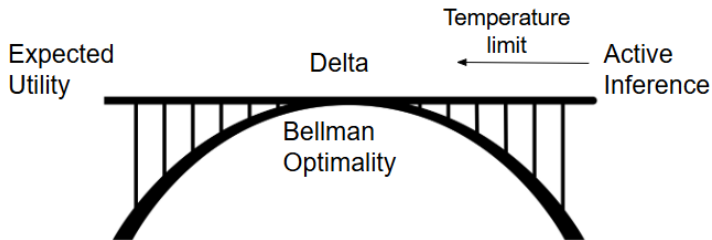
From Active Inference to Exp. Utility

Reward Maximization Through Discrete Active Inference (2023)

$$\text{Softmax Preference Distribution: } P(s|C) = \frac{e^{\beta R(s)}}{\sum_s e^{\beta R(s)}} \quad (1)$$
$$\longrightarrow \lim_{\beta \rightarrow +\infty}$$



From Active Inference to Exp. Utility



Risk Aversion

Information Sensitivity



Bernoulli 1713

Exhibit Information Sensitivity

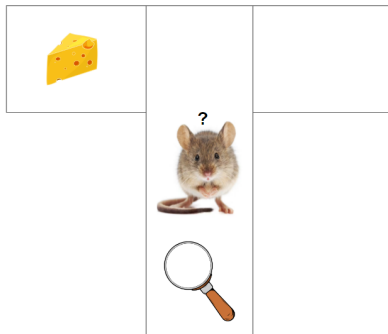


Figure 4: T-Maze POMDP ^{2 3}

$$U(R) = R(s)^c : c \in \mathbb{R}_+$$

$$\text{Risky strategy: } E[U(L_1)] = 0.5 \cdot 0 + 0.5 \cdot 1^c$$

$$\text{Risk averse strategy: } E[U(L_2)] = 1^c$$

$$E[U(L_2)] \succ E[U(L_1)] \text{ iff. } c < 1$$

(2)

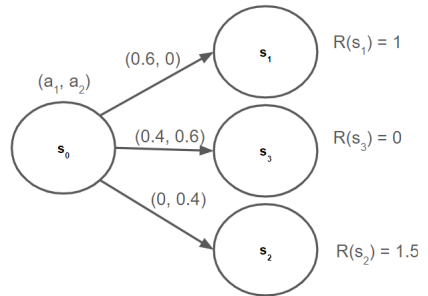
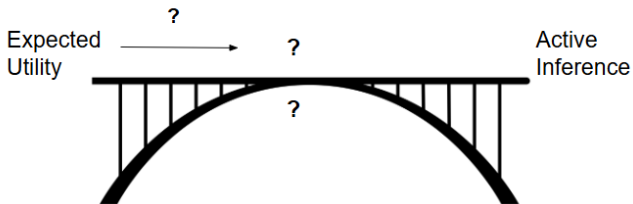


Figure 5: Paraglider POMDP

From EU to Active Inference



Information-Theoretic Constraint



Figure 6:
Cost: Large Action Space

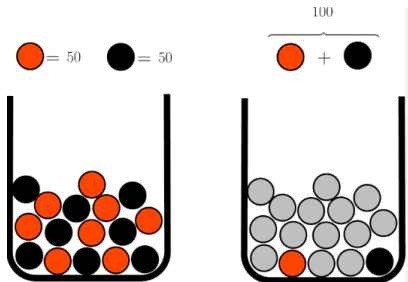
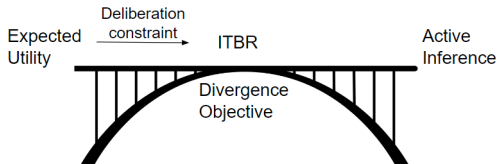


Figure 7:
Confidence: Model Uncertainty

$$\arg \min_{Q(s|a) \in \Delta(s|a)} D_{KL}[Q(s|a) || P(s)] \quad (3)$$

Connection: Divergence Objective ⁵



→ Maximize Utility and minimize shift in Beliefs ⁶

ITBR Lagrangian optimization problem:

$$\begin{aligned} F_{ITBR}(Q) &= \sum_s Q(s|a) \left(U(s, a) - \frac{1}{\beta} \log \frac{Q(s|a)}{P(s|a)} \right) \\ &= \underbrace{\sum_s Q(s|a) U(s, a)}_{\text{Max. Expected Utility}} - \underbrace{\sum_s Q(s|a) \left(\frac{1}{\beta} \log \frac{Q(s|a)}{P(s|a)} \right)}_{\text{Min. Shift}} \end{aligned} \quad (4)$$

The solution to which is the Softmax distribution:

$$Q^*(s|a) \propto P(s|a) \cdot e^{\beta U(s, a)} := P^*(s) \quad (5)$$

which is plugged into the objective function (1) to obtain the divergence objective⁵.

Optimal agency for ITBR and active inference can be reformulated

$$\begin{aligned} \text{Divergence objective: } a^* &= \arg \min_{a \in \mathbb{A}} D_{KL}[P(s_\tau | a_t) || P^*(s)] \\ \implies \arg \min_{a \in \mathbb{A}} (-F_{ITBR}) &= \arg \min_{a \in \mathbb{A}} G \end{aligned} \tag{6}$$

The Lagrangian and preference distribution derived analogously in a POMDP:

$$F_{ITBR}(Q) = \sum_s Q(o, s | a) \left(U(o, s, a) - \frac{1}{\beta} \log \frac{Q(o, s | a)}{P(o, s | a)} \right) \quad (7)$$

$$\implies \text{Divergence Obj. : } a^* = \arg \min_{a \in \mathbb{A}} D_{KL}[P(o_\tau, s_\tau | a_t) || P^*(o, s)]$$

Whereby expected free energy G and $-F_{ITBR}$ differ - incentivizing informative observations:

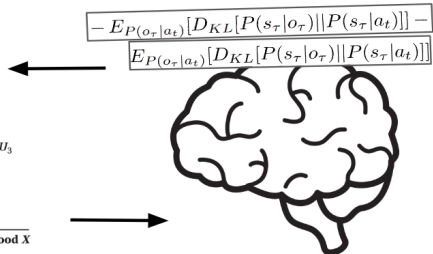
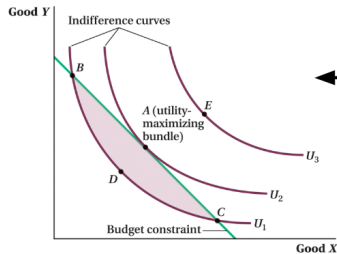
$$G - E_{Q(o|a)} H[Q(o|s)] = -F_{ITBR} \quad (8)$$

And the extrinsic value differs as follows ⁵:

$$P^*(s) = \frac{e^{U(R(s))}}{\sum_s e^{U(R(s))}} \quad \text{and} \quad P(s|C) = \frac{e^{R(s)}}{\sum_s e^{R(s)}} \quad (9)$$

Conclusion

- Bridge: Expected utility $\rightarrow$ ITBR $\rightarrow$ Active inference
- Key differences for POMDPs (5) - Which free energy?
- Potential for active inference economies ⁷



END

Addressing the Subsumption Thesis: A Formal
Bridge between Microeconomics and Active
Inference

Slides available on
<https://github.com/noeromeo>

QUESTIONS?

Citations:

- 1: Da Costa & Tenka & Zhao & Sajid, 2024, pg. 2
- 2: Da Costa, Friston, Parr, Sajid, 2022, pg. 10
- 3: Da Costa & Tenka & Zhao & Sajid, 2024, pg. 9
- 4: Da Costa & Tenka & Zhao & Sajid, 2024, pg. 10
- 5: Millidge & Buckley, 2021
- 6: Ortega, Braun, Dyer, Kim, Tishby, 2015
- 7: Hyland, Gavenciak, Da Costa, et al., in preparation

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<https://doi.org/10.1016/j.physrep.2023.07.001>

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Appendix

Tangent: Preferences in Active Inference

Consider the following example: (NOT PARAMETRIZED C)

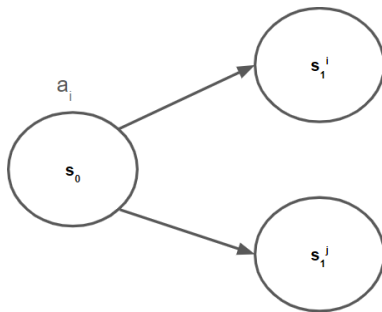


Figure 8: Transition between states given some action

$$P(s_1^i|a_1), P(s_1^j|a_1) = 0.6, 0.4 \quad P(s_1^i|a_2), P(s_1^j|a_2) = 0.1, 0.9$$

$$P(s_1^i) = 0.6 \cdot P(a_1) + 0.1 \cdot P(a_2) = 0.1$$

$$\rightarrow P(s_1^i) \leq 0.6$$

Active inference objective functions

$$\begin{aligned}\text{MDP EFE: } G_{\tau}(a_t) &= D_{KL}[P(s_{\tau}|a_t, s_t)||P(s|C)] \\ &= - \underbrace{\mathfrak{H}[P(s_{\tau}|a_t, s_t)]}_{\text{Entropy of future states}} - \underbrace{E_{P(s_{\tau}|a_t)}[\log P(s|C)]}_{\text{Expected Surprise}}\end{aligned}$$

$$\begin{aligned}\text{POMDP EFE: } G_{\tau}(a_t) &= \underbrace{E_{P(s_{\tau}|o_t, a_t)}\mathfrak{H}[P(o_{\tau}|s_{\tau})]}_{\text{Ambiguity}} + \underbrace{D_{KL}[P(s_{\tau}|a_t, o_t)||P(s|C)]}_{\text{Risk}} \\ &= - \underbrace{E_{P(o_{\tau}|a_t)}[D_{KL}[P(s_{\tau}|o_{\tau})||P(s_{\tau}|a_t)]]}_{\text{Intrinsic Value}} - \underbrace{E_{P(o_{\tau}, s_{\tau}|a_t)}[\log P(o|C)]}_{\text{Extrinsic Value}}\end{aligned}\tag{10}$$

2nd Exhibit

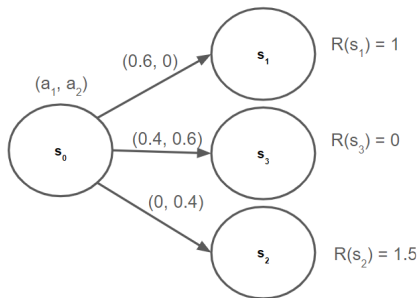


Figure 9: Paraglider POMDP

Desiderata⁴: Risk Aversion

Paraglider calc

Expected Utility and Active Inference for the 'Paraglider' MDP

The single-step MDP is specified as follows. Therefore note that the subscript does not pertain to the period:

$$\mathbb{S} = s_1, s_2, s_3$$

$$\mathbb{A} = a_1, a_2$$

$$\{P(s_1|a_1), P(s_2|a_1), P(s_3|a_1)\} = \{0.6, 0, 0.4\}$$

$$\{P(s_1|a_2), P(s_2|a_2), P(s_3|a_2)\} = \{0, 0.4, 0.6\}$$

$$\{R(s_1), R(s_2), R(s_3)\} = \{1, 1.5, 0\}$$

Consider an expected utility agent with utility function $U(R(s)) = R(s)^c$ where $c \in \mathbb{R}^+$. As such,

$$E[U(a_1)] = 0.6 \cdot 1^c$$

$$E[U(a_2)] = 0.4 \cdot 1.5^c$$

$$\text{For } c < 1 \rightarrow \arg \max_{a \in \mathbb{A}} E[U(a)] = a_1$$

$$\text{For } c > 1 \rightarrow \arg \max_{a \in \mathbb{A}} E[U(a)] = a_2$$

So a risk-averse expected utility agent will scale the smaller but safer mountain.

The active inference agent however is indifferent between the two actions. If we assume the preference distribution to be a softmax on the rewards, then we can ignore the normalizing denominator as it is constant w.r.t to action. Therefore we can write the relevant objective function as:

$$G(a_t) = - \sum_s P(s_t|a_t) \cdot R(s_t) - \sum_s P(s_t|a_t) \log \frac{1}{P(s_t|a_t)}$$

$$G(a_1) = -0.6 - 0.3065 - 0.3066 = G(a_2)$$

$$\rightarrow \arg \min_{a \in \mathbb{A}} G(a) = \{a_1, a_2\}$$

Therefore the optimal action of the risk-averse expected utility agent is a subset of the optimal active inference agency.

(9)

$$P^*(s|a) = \frac{P(s|a)e^{\beta U(s)}}{Z_\beta}$$

$$\frac{1}{\beta} \ln(P^*(s|a) \cdot Z_\beta) = U(s)$$

plug in for $U(s)$ (7)

$$\begin{aligned} & \operatorname{argmax}_a \frac{1}{\beta} E_{Q(s|a)} [\ln P^*(s|a) + \ln(Z_\beta)] - \frac{1}{\beta} \ln \frac{Q(s|a)}{P(s|a)} \\ &= \operatorname{argmax}_a E_{Q(s|a)} [\ln P^*(s|a) + \ln(Z_\beta) - \ln Q(s|a) + \ln P(s|a)] \\ &= \operatorname{argmin}_a E_{Q(s|a)} [-\ln P^*(s|a) - \ln(Z_\beta) + \ln Q(s|a) - \ln P(s|a)] \\ & \quad \operatorname{argmin}_a D_{\text{KL}}[Q(s|a) || P^*(s|a)] \end{aligned}$$

→ (10)

(13.5)

$$P^*(o, s|a) = \frac{P(o, s|a)e^{\beta U(o, s, a)}}{Z\beta}$$

plug in 13

end

$$= \operatorname{argmin}_a D_{KL}[Q(o, s|a) || P^*(o, s|a)]$$